25/2/2026 EXP: NO:- 6

# VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM:-

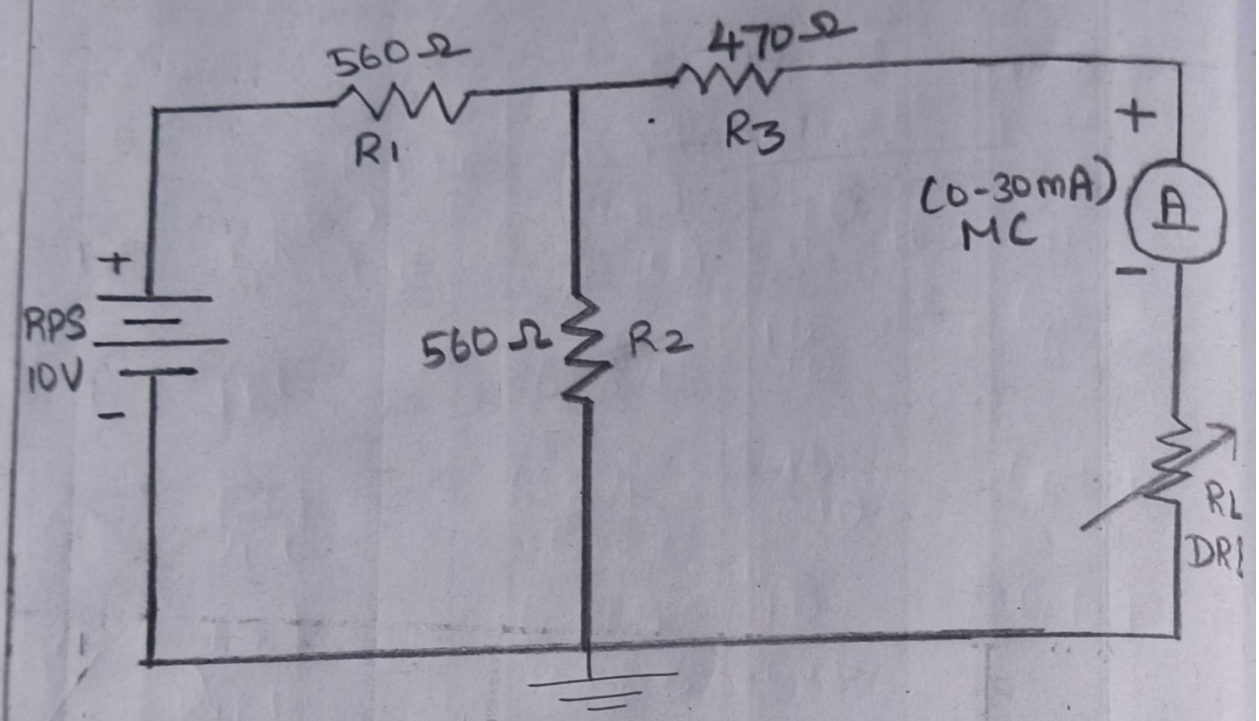
## AIM:-

To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance.

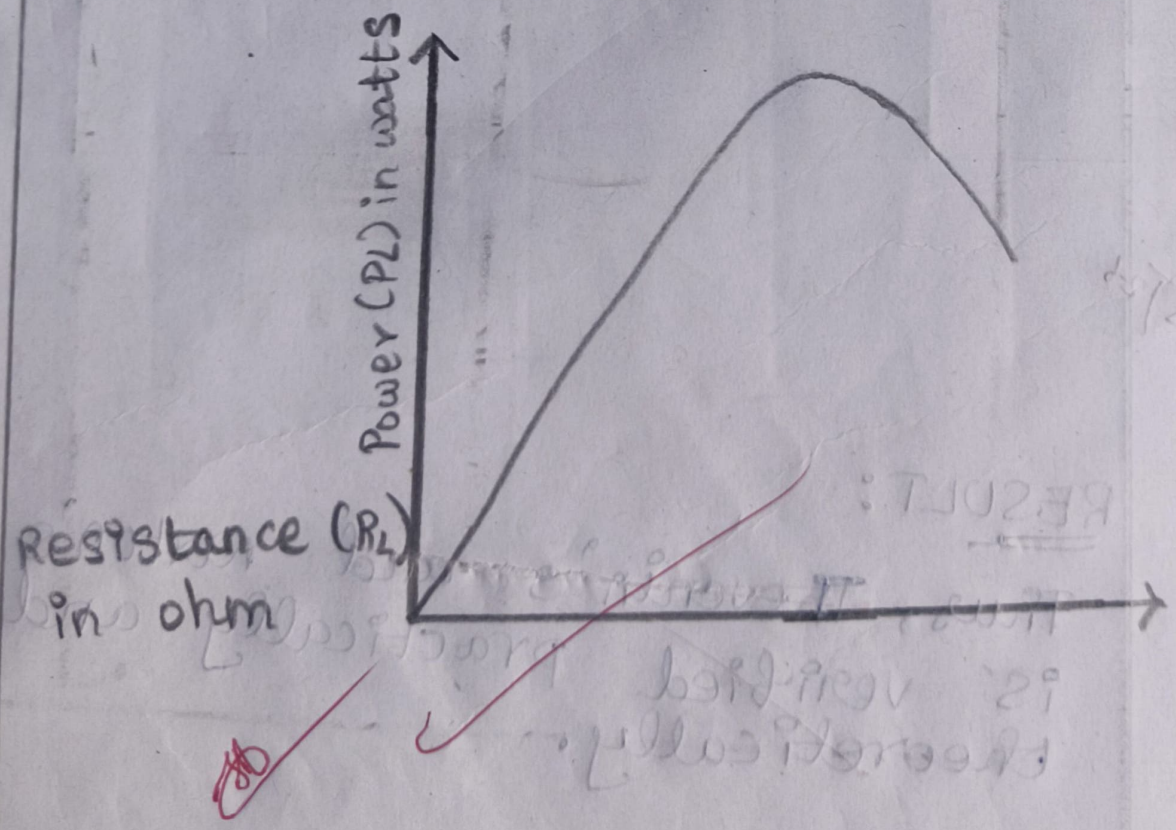
## APPARATUS REQUIRED:-

| S.NO | Name of the apparatus | Range/Rating | Quantity |
|---|---|---|---|
| 1. | Voltmeter | (0-15 V) MC | 1 |
| 2. | Ammeter | (0-500 mA) MC | 1 |
| 3. | Resistors | 560 Ω, 470 Ω | 2, 1 |
| 4. | RPS (DC Supply) | 15V | 1 |

## Circuit Diagram: For Maximum power Transfer Theorem:-




## MODEL GRAPH:-



TABULATION:-

TABLE (1): FOR THEORETICAL CALCULATION : $R_{th} = 750\,\Omega$

| S.NO | Load Resistance ($R_L$) in ohm | Load current ($I_L$) in mA | Load voltage ($V_L$) in volts | Load power ($P_L$) in watt |
|---|---|---|---|---|
| 1. | 1K | 6.84 | 10 | 68.4 |
| 2. | 47 | 6.82 | 10 | 68.2 |
| 3. | 1M | 6.83 | 10 | 68.3 |
| 4. | 15K | 6.78 | 10 | 67.9 |

O/P verified

PROCEDURE :-

1. Make connection as per the circuit diagram.
2. Change the resistors $R_L$ whose value close to $R_{th}$, measure the corresponding $V_L$, $I_L$ and calculate the power $P_L$ and enter into the table (2).
3. Plot a graph between $R_L$ and $P_L$ and find the $R_L$ corresponding to maximum power transfer.
4. Verify the measured values of $R_L$ at maximum power transfer as same as calculated and found graphically.

TABLE (2): FOR PRACTICAL CALCULATION:-

| S.NO | Load Resistance ($R_L$) in ohm | Load current ($I_L$) in amps | Load voltage ($V_L$) in volts | Load power ($P_L$) in watts |
|---|---|---|---|---|
| 1. | 1 K | 6.85 | 10 | 68.5 |
| 2. | 47 | 6.83 | 10 | 68.3 |
| 3. | 1 M | 6.84 | 10 | 68.4 |
| 4 | 15 K | 6.79 | 10 | 67.9 |

Op. verified

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MODEL CALCULATION:-

Given:

$R_1 = 560\,\Omega$

$R_2 = 560\,\Omega$

$R_3 = 470\,\Omega$

Supply voltage $= 10V$

Step 1: find $R_{Th}$:

$$\Rightarrow R_{Th} = R_3 + \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

MARKS SPLIT-UP :-

| SL.NO | Description | Marks Allotted | Marks obtained |
|---|---|---|---|
| 1. | Experimental setup/ connections | 20 marks | 20 |
| 2. | observation/ Readings | 20 marks | 20 |
| 3. | Calculations/ Results | 20 marks | 10 |
| 4. | Viva | 20 marks | 20 |
| 5. | Record | 20 marks | 20 |
| | | Total | 90 |

$$= 470 + \left(\frac{560 \times 560}{560 + 560}\right)$$

$$= 470 + 280$$

$$= 750\,\Omega$$

Step 2: find $V_{Th}$

$$V_{Th} = 10 \times \frac{R_2}{R_1 + R_2} = \frac{10 \times 560}{560 + 560}$$

$$= 5V$$

Step 3: $R_L = 750\,\Omega$

$R_L = R_{Th}$

$R_L = 750$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{5}{760 + 760} =$$

$$= 0.0032\,A$$

$$\Rightarrow V_L = I_L \times R_L$$

$$= 2.5V$$

$$\Rightarrow R_L = V_L + I_L$$

$$= 8.33\,\Omega$$

90

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RESULT:-

Thus, maximum power transfer theorem is verified practically and theoretically.